

Embodied Ink

A Multisensory Reinterpretation of Chinese Calligraphy Through Digital Twins and Immersive Realities

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Abstract

For thousands of years, Chinese culture has regarded calligraphy as both a form of painting and a deeply physical, philosophical practice. While traditionally manifesting as a two-dimensional art form, Chinese calligraphy emerges from three-dimensional, embodied movements infused with breath, emotion, and intention. This demo/video paper introduces Embodied Ink, a multimedia interactive installation that reinterprets Chinese calligraphy through motion capture and generative AI. By translating the audience movements into real-time dynamic visuals and soundscapes, the project reveals the hidden kinetics and philosophical depth underlying calligraphy. Embodied Ink reimagines the static medium of calligraphy as a dynamic interplay of forces and particles, inviting viewers to experience calligraphy as a living, ever-changing art form.

CCS Concepts

• Human-centered computing → Interaction design → Interaction design process and methods → Activity centered design

Keywords

Chinese calligraphy, motion capture, generative AI, multimedia installation

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1 Introduction

Chinese calligraphy, the artful writing of Chinese characters, seamlessly blends visual expression with the interpretation of literary meaning. Highly esteemed across East Asia, this tradition has been widely practiced in China for centuries. Though its final form appears as a visual art on scrolls, Chinese calligraphy transcends mere representation, embodying a profound physical and philosophical practice. It originates from intricate three-dimensional movements of the body, infused with emotion and shaped by the dynamic interaction between the artist, brush, and medium.

Embodied Ink reimagines this timeless practice by integrating contemporary digital technologies. Through motion capture and interactive installations, the project uncovers the hidden dynamics of calligraphic creation, transforming it into a living, multisensory experience that bridges ancient tradition and modern innovation. [1,2]



Figure 1 Embodied Ink Installation

2 SYSTEM DESIGN

This system design diagram illustrates the conceptual and technical framework of an interactive installation inspired by Chinese calligraphy. The workflow is divided into two interconnected streams, Visual and Music, that converge into a unified immersive installation. At its foundation, the system

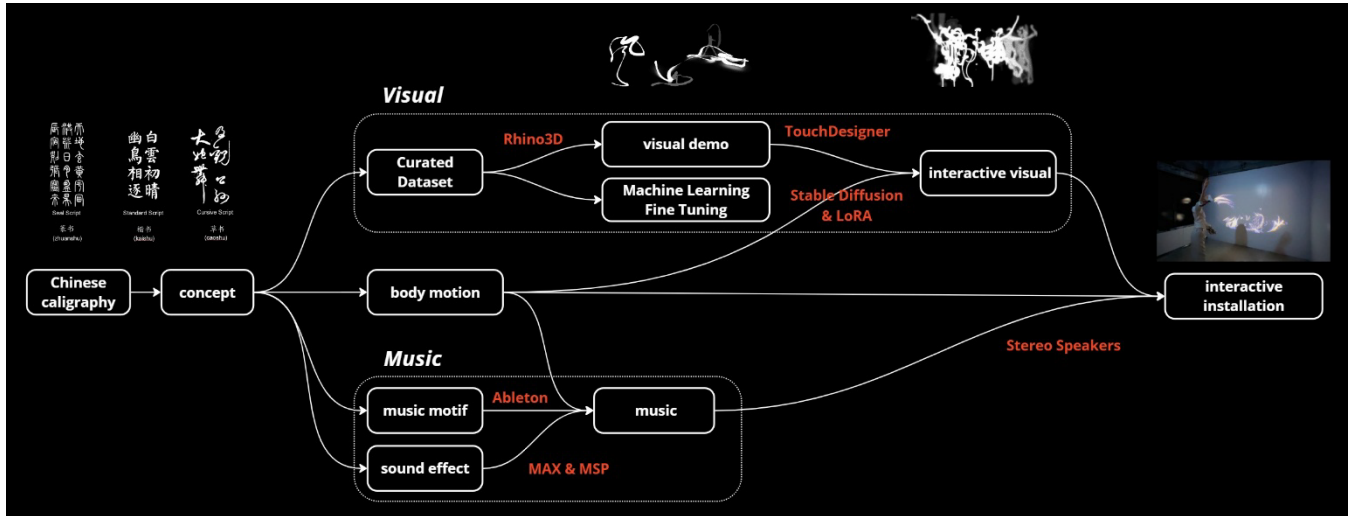


Figure 2 Project System Design

draws inspiration from Chinese calligraphy, specifically the three major styles: Seal Script (Zhuanshu), Standard Script (Kaishu), and Cursive Script (Caoshu). These historical forms serve not only as aesthetic references but also as cultural and philosophical anchors for the project. [1,2]

From the conceptual core, the design expands into the Visual pathway. A curated dataset of calligraphy strokes, movements, and compositions is developed. This dataset is processed through two directions: (1) Rhino3D [3], which enables the generation of 3d visual demonstrations and structural modeling of strokes, and (2) Machine Learning Fine-Tuning, using tools such as Stable Diffusion [4] and LoRA [5] to synthesize calligraphy-inspired imagery. These outputs are integrated within TouchDesigner [6], where they are transformed into interactive experiments and visual artworks. The visuals respond to real-time data and audience motion input, creating a fluid and dynamic expression of calligraphic aesthetics within a digital environment. The system bridges traditional artistry with computational creativity, resulting in a new form of visual art that extends beyond static brushstrokes to embodied, performative visuals.

Parallel to the visual pipeline, the Music stream contributes to the multisensory experience. It begins with the generation of music motifs and sound effects, which are composed and modulated in Ableton [7]. MAX & MSP [8] is incorporated to handle signal processing and interactivity, ensuring that sound is not only an accompaniment but an active participant in shaping the immersive atmosphere. The music is designed to resonate with the rhythmic qualities of calligraphy, echoing the brush's tempo, intensity, and fluidity. Through stereo speaker output, sound interacts with visuals, enveloping the audience in a cohesive environment.

The system design emphasizes hybridity: ancient Chinese calligraphy meets cutting-edge AI, while visual and musical modalities intertwine to augment human perception. This hybridity resonates with broader media art discourses on software

as cultural form [1] and radical materiality in interaction design [9].

3 Conclusion

Embodied Ink expands the legacy of calligraphy into contemporary discourse, offering a framework for reanimating intangible heritage through immersive, computational media. By merging motion capture and generative AI, it dissolves boundaries between choreography, computation, and visual art. This work invites viewers to experience calligraphy not as a static artifact but as a living, ever-evolving art form, bridging ancient traditions and modern technologies [1,2].

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